

Luminous Atmospheres 360: Comparing VR HMDs and UHD Screens for Reproducing Architectural Daylight Atmospheres

Michèle Atié*
Nantes Université, ENSA
Nantes, Ecole Centrale
Nantes, CNRS,
AAU-CRENAU, UMR 1563
Nantes, France

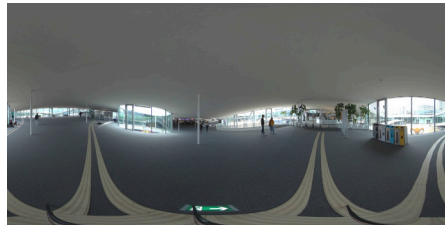
Céline Drozd†
Nantes Université, ENSA
Nantes, Ecole Centrale
Nantes, CNRS,
AAU-CRENAU, UMR 1563
Nantes, France

Toinon Vigier‡
Nantes Université, Ecole
Centrale Nantes, LS2N,
UMR 6004 Nantes, France

Daniel Siret§
Nantes Université, ENSA
Nantes, Ecole Centrale
Nantes, CNRS,
AAU-CRENAU, UMR 1563
Nantes, France



Bibliothèque Sainte-Geneviève, Paris



Rolex Learning Center, Lausanne

Impressions related to Bodily Sensations

Buildings	Real	BDC			Real	ELR			Real	ESP			Real	MLL			Real	PDJ			Real	RLC		
		Mean	SD	CI		Mean	SD	CI		Mean	SD	CI		Mean	SD	CI		Mean	SD	CI		Mean	SD	CI
In this place I feel...	Mean	2.33	0.14	2.05	2.19	1.81	1.75	2.83	2.01	1.85	2.27	1.88	2.15	2.28	2.14	2.00	1.70	1.45	2.19	2.48	2.48	2.30	2.48	2.30
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.500	0.849	0.849	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
light	Mean	1.30	1.48	1.85	1.24	1.48	1.85	1.00	1.57	1.70	1.23	1.88	1.08	1.71	2.45	1.30	1.88	1.08	1.28	1.38	1.28	1.38	1.28	1.38
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
in a state of fullness and peace	Mean	2.28	2.28	2.28	2.16	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
disoriented	Mean	1.01	1.27	1.35	1.14	1.40	1.20	1.08	1.14	1.20	1.21	1.78	1.20	1.25	1.48	1.70	1.28	1.57	2.05	1.48	1.57	2.05	1.48	1.57
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
hypercalibrated	Mean	1.70	2.14	1.85	1.24	2.00	2.00	1.75	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85	1.85
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
invited to meditate	Mean	2.14	2.14	1.75	1.28	2.07	2.07	2.04	2.48	2.30	2.41	1.88	2.30	2.29	1.71	1.80	1.80	1.78	1.40	1.57	1.87	1.87	1.87	1.87
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
off balance	Mean	1.88	1.88	1.88	1.14	1.78	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
in retreat	Mean	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84	1.84
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
in contemplative stillness	Mean	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
lost	Mean	1.05	1.78	1.20	1.25	1.25	1.25	1.17	1.14	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
absorbed by sensory, calm and clarity	Mean	2.30	2.30	2.30	2.11	2.02	2.02	2.15	2.20	2.40	2.02	1.78	2.02	2.02	2.02	2.02	2.02	2.02	2.02	2.02	2.02	2.02	2.02	2.02
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
in a bath of light	Mean	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
suspended in inebriation	Mean	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151
on an endless circular path	Mean	1.80	2.00	2.00	1.87	1.40	1.40	1.00	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40
	p-value	0.308	0.500	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849	0.849
	context d	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151	0.151

Figure 1: Examples of two 360° photographs projected in the VR HMD (on the left) and an excerpt of results showing the differences in perception between each display and the real condition (VR HMD vs real; VR HMD vs screen) for each building (on the right).

ABSTRACT

This paper studies the potential of VR HMDs to replicate subjective impressions of luminous atmospheres using 360° photographs. We compare the perception of luminous atmospheres in two settings—VR HMD and UHD screen—against real iconic buildings. Evaluations were conducted through three distinct experiments: in seven real indoors of iconic buildings, using 360° photographs of these indoors viewed in a VR HMD, and using 2D images from these indoors displayed on a UHD screen. Results indicate that VR HMDs are a promising tool for studying luminous atmospheres, providing a closer perception of real-world conditions.

Index Terms: Perception, virtual reality head-mounted-display (VR HMD), screen, fidelity, luminous atmospheres, education.

1 INTRODUCTION

Representing indoor scenes from reference buildings is a common practice in architectural design education, particularly for studying luminous atmospheres. This approach enables students to analyze how light influences perception and subjective impressions. Con-

ventional methods such as 2D photographs presented in books or on display screens are used to depict these reference buildings and their luminous atmospheres. However, these methods are limited by their inability to provide a full field of view, making it challenging to convey the complexity of an atmosphere, which requires spatial engagement to be fully understood [1]. Recent advancements in technology have introduced new tools for indoor representations, including immersive displays like virtual reality (VR) head-mounted displays (HMDs). These devices have become increasingly prevalent in architectural education, enabling the exploration of diverse environments across multiple locations [3]. Immersive scenes projected in VR HMDs have demonstrated their potential to convey impressions of daylight environments, such as visual comfort [2] in standard spaces like offices. However, little research has focused on the ability of immersive photographs to represent complex subjective impressions of luminous atmospheres of real iconic architectural buildings.

Thus, we aim to evaluate the potential of a VR HMD in conveying luminous atmospheres from 360° photographs compared to conventional displays, such as ultra-high definition (UHD) screens

Henri Labrouste's "Bibliothèque Sainte-Geneviève" (BSG); and SANAA's "Rolex Learning Center" (RLC).

We conducted experiments in the seven indoors, using three distinct setups: directly within the real spaces, through 360° photographs of these scenes projected in a VR HMD, and through viewports of these scenes displayed on a UHD screen. The methodology for this study involved assessing luminous atmospheres through a specifically designed questionnaire administered across the three settings. The questionnaire included 70 subjective impressions, rated on a scale from 1 to 3, that examined five dimensions of luminous atmospheres: bodily sensations (e.g., in this place I feel light), symbolic references (e.g., this place reminds me of a ship), atmosphere (e.g., the atmosphere in this place is mysterious), space (e.g., the space seems empty), and light (e.g., I see a foggy effect). These impressions were derived from expert articles using a method that analyzed descriptive terms in architectural texts. The guidelines, defined as codes specifying the meaning of expressions (e.g., for light, everything related to "In this place, I see..."), were collaboratively validated by research architects.

2.1 Real Experiments

In real experiments, we have developed a method to assess the perception of luminous atmospheres in the seven buildings. Participants on-site evaluated the questionnaire from a specific viewpoint in each building, with groups per building: MLL (24), BDC (45), RLC (41), ESP (42), BSG (38), ELR (12), PDJ (21). Prior to these evaluations, we captured 360° HDR stereoscopic panoramas using an "Insta360 Pro 2" camera, which records nine multiple exposures through six fisheye lenses, under a stereoscopic capture. Subsequently, luminance measurements were conducted across different points of the scene using a luminance meter.

2.2 VR HMD Experiment

The captured images were processed using the camera's software, which exported stereoscopic images from multiple exposures. These were merged into 32-bit HDR format and subsequently converted into 8-bit format after applying the Drago tone-mapping operator (TMO). The images' luminance values were then verified against real-world measurements. A virtual interactive environment was created in Unity 3D using an HTC Vive Pro 2, where 21 participants evaluated a questionnaire displayed for each of the seven scenes, which were captured in real buildings and presented in a randomized order.

2.3 UHD Screen Experiment

The images used in the virtual application were also employed in the screen-based experiment. For each scene, two 2D images with a 3:2 aspect ratio were extracted and projected for evaluation. In this experiment, 20 participants evaluated the images in a randomized order, on a computer screen positioned in front of them. During each evaluation session, the two images of each panoramic scene were projected onto a more distant UHD screen with a resolution of 3840 x 2160 within the room. Both screens were calibrated to the BT.709 color standard with a luminance of 200 cd/m².

3 STATISTICAL ANALYSIS

We aim to find which display—VR HMD or screen—most closely replicates the perception of reality by comparing the evaluations between the settings (Reality vs. VR HMD, and Reality vs. screen). We used the non-parametric Wilcoxon rank-sum test to analyze ordinal data and determine significant differences between conditions (p-value). Additionally, we calculated Cliff's delta effect size to evaluate how specific displays exaggerated or diminished subjective impressions compared to their assessment in reality. We present the results using the following color-coded scheme: gray indicates no significant differences between reality and a specific display;

light blue indicates a slight diminishment of the impression compared to reality; dark blue indicates a strong diminishment of the impression; light red indicates a slight exaggeration of the impression compared to reality, and dark red indicates a high exaggeration of the impression compared to reality (Figure 1).

4 RESULTS

Overall, most participants lacked experience with real buildings in VR HMD or on screens, except for the PDJ building (12 participants had seen it once or multiple times in VR and 6 on screen). In the VR experiment 54% had tried the VR HMD at least once or a few times. The average age of participants varied between 35.0 and 42.7 for most conditions, including screen (35.0, sd: 12.9), VR (32.3, sd: 14.6), ESP (32.7, sd: 21.8), BDC (37.3, sd: 15.6), PDJ (37.7, sd: 11.0), ELR (38.2, sd: 16.7), and MLL (42.7, sd: 18.1). However, differences were observed with younger averages in RLC (20.4, sd: 1.7) and BSG (23.7, sd: 9.2). We present an excerpt of our results, focusing on the dimension related to bodily sensations. Fewer differences were observed between the VR HMD and real conditions compared to the screen, where significant discrepancies were noted between certain cases—for example, the impression "*In this place I feel crushed*" was exaggerated across all locations in the screen. Similarly, "*In this place, I feel hypnotized*" was highly exaggerated in the screen for ELR and RLC, while "*In this place, I feel off-balance and lost*" was exaggerated in the screen for ELR, ESP, MLL, and RLC. Nonetheless, a few impressions remained perceptibly similar across both devices, though neither fully replicated the real condition. For example, "*In this place, I feel light*" was slightly diminished in both displays for BSG and highly diminished for ELR. Finally, in a few rare instances, the screen provided a better reproduction of impressions such as "*In this place I feel invited to meditation*" for PDJ.

5 CONCLUSION AND FUTURE WORK

This work presented preliminary results based on an analysis of 70 subjective impressions, showing the differences between seven real buildings and two displays: VR HMD and UHD screen. The findings suggest that the VR HMD reproduces overall real impressions more effectively, offering a promising level of fidelity compared to the UHD screen. However, certain impressions were found to be more effectively conveyed through the UHD screen, while others remained perceptibly similar across both devices, suggesting that future studies should explore the reasons behind these differences. Future research should apply this method in an educational context and explore a broader range of indoor scenes and varying lighting conditions.

ACKNOWLEDGMENTS

The authors would like to thank Maxim Spur for helping in the statistical analysis and the VR development, "XP-LAB, Halle 6 Ouest, Nantes" for recruiting participants and conducting the screen experiment, Meriem Belghoul and Niren Cunthen for their contribution in the collection of data in all real buildings.

REFERENCES

- [1] G. Böhme. The theory of atmospheres and its applications (translated by a.-chr. engels-schwarzpaul). *Interstices: Journal of Architecture and Related Arts*, pp. 93–100, 2014. 1
- [2] S. Rockcastle, M. Danell, E. Calabrese, G. Sollom-Brotherton, A. Mahic, K. Van Den Wymelenberg, and R. Davis. Comparing perceptions of a dimmable led lighting system between a real space and a virtual reality display. *Lighting Research & Technology*, 53(8):701–725, 2021. 1
- [3] P. S. Scarazzato, J. C. d. S. F. Matos, A. J. G. L. França, and T. D. Pavani. Challenges in lighting education: a recommended practice. In *IOP Conference Series: Earth and Environmental Science*, vol. 1099, p. 012040. IOP Publishing, 2022. 1